A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Chemical equilibria

At AS you predicted which way an equilibrium moves. Now you put numbers on it. The equilibrium constants K_c and K_p measure exactly where an equilibrium lies, you can calculate them from experimental data, and they explain why changing the temperature behaves so differently from changing concentration, pressure or adding a catalyst.

The equilibrium constant K_c

In a closed system a reversible reaction settles into **dynamic equilibrium**: the forward and backward reactions are still happening, but at the same rate, so the concentrations of everything stop changing. For a given reaction at a given temperature, the equilibrium concentrations always fit one fixed ratio, whatever amounts you started with. That ratio is the equilibrium constant.

For the general reaction $mA + nB \rightleftharpoons pC + qD$:

$$K_c = \frac{[C]^p\,[D]^q}{[A]^m\,[B]^n}$$

- **Square brackets** mean the concentration at equilibrium, in mol dm^{-3}.
- **Products go on top**, reactants underneath, and each concentration is raised to the power of its number in the balanced equation.
- **K_c belongs to an equation as written.** Write the equation backwards and the new K_c is the reciprocal of the old one; double every number in the equation and K_c is squared. Always quote the equation with the value.

Homogeneous and heterogeneous equilibria

In a **homogeneous** equilibrium everything is in the same phase, for example all gases, or all dissolved in one solution. In a **heterogeneous** equilibrium there is more than one phase, such as a solid in contact with a gas or a solution. A pure solid (or a pure liquid that forms its own separate layer) has a fixed concentration however much of it is there, so it is left out of the K_c expression.

Equilibrium	Type	K_c
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	homogeneous	$\frac{[HI]^2}{[H_2][I_2]}$
$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$	homogeneous (one liquid mixture)	$\frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]}$

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	heterogeneous	$[CO_2]$
$Ag^+(aq) + Fe^{2+}(aq) \rightleftharpoons Ag(s) + Fe^{3+}(aq)$	heterogeneous	$\dfrac{[Fe^{3+}]}{[Ag^+][Fe^{2+}]}$

COMMON MISTAKE

Water is left out only when it is the solvent, present in huge excess. In the esterification above, water is one of four liquids mixed in a single solution and is formed in similar amounts to the ester, so it **does** appear in K_c.

Working out the units of K_c

K_c has units only when the powers on the top and bottom do not cancel. Put mol dm^{-3} in for every concentration and simplify.

Units of K_c for N₂(g) + 3H₂(g) ⇌ 2NH₃(g)

1. Write the expression: $K_c = \dfrac{[NH_3]^2}{[N_2][H_2]^3}$.
2. Substitute the unit: $\dfrac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})^3} = \dfrac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})^4}$.
3. Cancel: $(\text{mol dm}^{-3})^{-2} = \text{mol}^{-2}\,\text{dm}^6$.

A quick check: there are 2 molecules on top and 4 underneath, so the unit is (mol dm^{-3})$^{2-4}$.

For $H_2 + I_2 \rightleftharpoons 2HI$ there are 2 on top and 2 underneath, so K_c has **no units**. For $2SO_2 + O_2 \rightleftharpoons 2SO_3$ it is (mol dm^{-3})$^{2-3}$ = dm^3 mol^{-1}.

Calculating K_c from experimental data

A question usually gives the starting amounts of the reactants and one piece of information about the equilibrium mixture: the amount of one substance, or the percentage of a reactant that has reacted. The balanced equation then tells you how everything else changed. Lay it out in a table with three rows, often called an ICE table: **I**nitial moles, **C**hange, **E**quilibrium moles. Then divide by the volume to get concentrations.

- **Reactant at equilibrium** = moles at the start − moles that reacted.
- **Product at equilibrium** = moles at the start + moles that formed.
- The changes are always in the ratio of the numbers in the balanced equation.

Example 1: H₂ + I₂ ⇌ 2HI (no units)

1. 0.500 mol of H_2 and 0.500 mol of I_2 are heated in a sealed 2.00 dm^3 flask. At equilibrium the flask contains 0.786 mol of HI. Find K_c.

2. HI formed = 0.786 mol. The equation says 2 HI form for every 1 H_2 and 1 I_2 used, so 0.393 mol of each reactant reacted.

3.

	H_2	I_2	HI
Initial / mol	0.500	0.500	0
Change / mol	−0.393	−0.393	+0.786
Equilibrium / mol	0.107	0.107	0.786
Concentration / mol dm^{-3}	0.0535	0.0535	0.393

4. $K_c = \dfrac{[\mathrm{HI}]^2}{[\mathrm{H_2}][\mathrm{I_2}]} = \dfrac{0.393^2}{0.0535 \times 0.0535} = 54.0$, with no units.

Because the units cancel, the volumes cancel too: putting the equilibrium moles straight in gives the same answer. That shortcut only works when K_c has no units.

Example 2: 2SO₂ + O₂ ⇌ 2SO₃ (with units)

1. 2.00 mol of SO_2 and 1.20 mol of O_2 are mixed in a 5.00 dm^3 vessel with a catalyst. At equilibrium there are 1.60 mol of SO_3. Find K_c.

2. 1.60 mol of SO_3 formed, so 1.60 mol of SO_2 and 0.800 mol of O_2 reacted (ratio 2 : 1 : 2).

3. Equilibrium moles: SO_2 2.00 − 1.60 = 0.400; O_2 1.20 − 0.800 = 0.400; SO_3 1.60.

4. Divide by 5.00 dm^3: $[SO_2]$ = 0.0800, $[O_2]$ = 0.0800, $[SO_3]$ = 0.320 mol dm^{-3}.

5 $$K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]} = \frac{0.320^2}{0.0800^2 \times 0.0800} = 200\ \text{dm}^3\,\text{mol}^{-1}$$

Here the volume does not cancel. Forgetting to divide by 5.00 would give 40.0, a wrong answer.

Practical: finding K_c for an esterification

Ethanoic acid and ethanol react slowly to make ethyl ethanoate and water. A little strong acid speeds the reaction up so that equilibrium is reached in about a week at room temperature:

$$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$$

Method

1. Use burettes to measure known volumes of ethanoic acid, ethanol and a dilute acid catalyst into a boiling tube. Stopper it, swirl and leave it for a week.
2. Find how much acid the catalyst alone contributes by titrating a separate sample of the same volume of catalyst with sodium hydroxide.
3. Wash the equilibrium mixture into a 250 cm^3 volumetric flask with distilled water, make it up to the mark and invert it to mix.
4. Pipette 25.0 cm^3 portions into conical flasks and titrate with standard sodium hydroxide using phenolphthalein, until you have concordant titres.

- **Starting amounts** of the acid and the alcohol come from their volumes: mass = density × volume, then moles = mass ÷ M_r.
- **The titration measures all the acid present**: the ethanoic acid left at equilibrium plus the catalyst. Subtract the catalyst's share to find the ethanoic acid.
- **Diluting with water does not spoil the result** because the titration is done quickly, and the uncatalysed reaction is far too slow to shift much in that time.
- **The pink end-point can fade**: as the hydroxide removes ethanoic acid, the equilibrium slowly shifts left, hydrolysing some ester and releasing more acid. Take the first permanent pink.
- **To check that a week was long enough**, set up identical mixtures and analyse them after different times. If K_c stops changing, equilibrium had been reached.

Calculating K_c from the titration

1 The mixture started with 0.200 mol of ethanoic acid and 0.200 mol of ethanol. Each 25.0 cm^3 portion of the diluted mixture needed 21.40 cm^3 of 0.500 mol dm^{-3} NaOH. The catalyst alone contains 0.0400 mol of H^+.

2. NaOH used per portion = 0.02140 dm^3 × 0.500 mol dm^{-3} = 0.01070 mol, so the 25.0 cm^3 portion held 0.01070 mol of acid.

3. The whole 250 cm^3 held ten times as much: 0.1070 mol of acid.

4. Ethanoic acid at equilibrium = 0.1070 − 0.0400 = 0.0670 mol.

5. Ethanoic acid that reacted = 0.200 − 0.0670 = 0.133 mol, so 0.133 mol of ethanol reacted, and 0.133 mol each of ester and water formed. Ethanol left = 0.200 − 0.133 = 0.0670 mol.

6. $K_c = \dfrac{0.133 \times 0.133}{0.0670 \times 0.0670} = 3.94$, with no units, so the total volume cancels.

   This ignores the water that came in with the dilute catalyst. Including it makes [H_2O] slightly bigger and K_c slightly bigger: a known limitation of the method.

Partial pressures and K_p

For reactions between gases it is often easier to measure pressures than concentrations, so a second equilibrium constant, K_p, is written with **partial pressures**.

- **Partial pressure** of a gas in a mixture: the pressure it would exert if it alone filled the whole volume of the mixture. It is written p_A or p(A).
- **Mole fraction**: $x_A = \dfrac{\text{moles of A}}{\text{total moles of gas}}$. The mole fractions of all the gases add up to 1.
- **Partial pressure** = mole fraction × total pressure: $p_A = x_A P$.
- **The partial pressures add up to the total pressure**: $P = p_A + p_B + p_C + \ldots$

For $mA(g) + nB(g) \rightleftharpoons pC(g) + qD(g)$:

$$K_p = \frac{p_C^{\,p}\, p_D^{\,q}}{p_A^{\,m}\, p_B^{\,n}}$$

Only gases appear in K_p. Solids, liquids and dissolved substances are left out, so for $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = p_{CO_2}$, with units of pressure. Units of K_p are worked out like those of K_c, using the pressure unit given in the question (atm or kPa): for $N_2 + 3H_2 \rightleftharpoons 2NH_3$, K_p is in atm^{-2} (or kPa^{-2}).

Example 3: N₂O₄(g) ⇌ 2NO₂(g)

1. 1.00 mol of N_2O_4 is allowed to reach equilibrium. 20.0% of it dissociates, and the total pressure is 1.00 atm. Find K_p.

2. N_2O_4 reacted = 0.200 mol, so NO_2 formed = 0.400 mol. At equilibrium: 0.800 mol N_2O_4 and 0.400 mol NO_2, 1.200 mol in total.

3. Mole fractions: $x_{N_2O_4} = 0.800/1.200 = \frac{2}{3}$; $x_{NO_2} = 0.400/1.200 = \frac{1}{3}$.

4. Partial pressures at 1.00 atm: $p_{N_2O_4} = \frac{2}{3} = 0.667$ atm, $p_{NO_2} = \frac{1}{3} = 0.333$ atm.

5. $$K_p = \frac{p_{NO_2}^2}{p_{N_2O_4}} = \frac{(1/3)^2}{2/3} = 0.167 \text{ atm}$$

   With the rounded values 0.333 and 0.667 you would get 0.166: another reason to keep full figures until the end.

Example 4: N₂(g) + 3H₂(g) ⇌ 2NH₃(g)

1. 2.00 mol of N_2 and 6.00 mol of H_2 reach equilibrium at a total pressure of 100 atm. The equilibrium mixture contains 1.00 mol of NH_3. Find K_p.

2. 1.00 mol of NH_3 formed, so 0.500 mol of N_2 and 1.50 mol of H_2 reacted. At equilibrium: N_2 1.50, H_2 4.50, NH_3 1.00; total 7.00 mol.

3. Partial pressures: $p_{N_2} = \frac{1.50}{7.00} \times 100 = 21.4$ atm; $p_{H_2} = \frac{4.50}{7.00} \times 100 = 64.3$ atm; $p_{NH_3} = \frac{1.00}{7.00} \times 100 = 14.3$ atm. Check: they add up to 100 atm.

4. $$K_p = \frac{p_{NH_3}^2}{p_{N_2}\, p_{H_2}^3} = \frac{14.3^2}{21.4 \times 64.3^3} = 3.58 \times 10^{-5} \text{ atm}^{-2}$$

   Keep the unrounded partial pressures in your calculator; rounding to three figures first can shift the last digit.

What changes K, and what does not

KEY IDEA

For a given equation, the value of K_c or K_p depends **only on temperature**. Changing concentration or pressure moves the **position** of equilibrium but leaves K unchanged, and a catalyst changes neither.

Concentration

Add more H_2 to an equilibrium mixture of H_2, I_2 and HI. For a moment the fraction $\frac{[HI]^2}{[H_2][I_2]}$ is smaller than K_c, because its bottom line has grown. The system is no longer at equilibrium. The forward reaction now runs faster than the backward one, using up H_2 and I_2 and making HI, until the fraction is back up to exactly K_c. The position has moved right; K_c has not changed.

Pressure

Increasing the total pressure changes every partial pressure at once, but not by the same overall factor on the top and bottom of K_p when the two sides have different numbers of gas molecules.

Why higher pressure gives more ammonia although K_p stays the same

1. Write each partial pressure as mole fraction × total pressure: $K_p = \dfrac{(x_{NH_3}P)^2}{(x_{N_2}P)(x_{H_2}P)^3}$.
2. Collect the P terms: $K_p = \dfrac{x_{NH_3}^2}{x_{N_2}\,x_{H_2}^3} \times \dfrac{1}{P^2}$.
3. Double P and the factor $1/P^2$ falls to a quarter. Straight after the change the right-hand side is smaller than K_p, so the mixture is not at equilibrium.
4. To restore K_p, the mole-fraction part must grow fourfold: more NH_3 forms and N_2 and H_2 are used up. The position shifts right, towards fewer gas molecules, and K_p ends where it started.

If both sides have the same number of gas molecules, the P terms cancel completely and pressure has no effect on the position at all.

Catalysts

A catalyst provides an alternative route with a lower activation energy for both the forward and the backward reaction. Both speed up by the same factor, so equilibrium is reached sooner but its position and the value of K are unchanged.

Temperature

Temperature is the one condition that changes K itself. If the forward reaction is **exothermic**, raising the temperature makes K **smaller**: the equilibrium mixture contains less product. If the forward reaction is **endothermic**, raising the temperature makes K **larger**. The new value of K then forces the concentrations to adjust, which is why the position also moves. The next section explains why, using entropy.

The size of K, extent of reaction and entropy

The bigger K is, the further the reaction goes before it reaches equilibrium. As a rough guide:

Value of K	What the equilibrium mixture is like
less than about 10^{-10}	essentially only reactants: the reaction does not happen to any noticeable extent
around 0.1	reactants predominate
around 1	similar amounts of reactants and products
around 10	products predominate
more than about 10^{10}	essentially only products: the reaction goes to completion

This links to Topic 12. The total entropy change of a reaction and its equilibrium constant are connected by

$$\Delta S_{\text{total}} = R \ln K$$

where R = 8.31 J K^{-1} mol^{-1}. A positive ΔS_{total} means K > 1, so products are favoured; a negative ΔS_{total} means K < 1. A reaction with K = 10^{10} has ΔS_{total} = 8.31 × ln 10^{10} = +191 J K^{-1} mol^{-1}. Since $\Delta G = -T\Delta S_{total}$, the same idea can be written $\Delta G = -RT \ln K$.

Why temperature changes K

1. $\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$, and $\Delta S_{\text{surroundings}} = -\frac{\Delta H}{T}$.
2. For an **exothermic** reaction ΔH is negative, so $\Delta S_{surroundings}$ is positive. As T rises, −ΔH/T gets **smaller**.
3. So ΔS_{total} becomes less positive (or more negative). Because ΔS_{total} = R ln K, ln K falls and K gets smaller: less product at equilibrium.
4. For an **endothermic** reaction $\Delta S_{surroundings}$ is negative, and rising T makes it less negative. ΔS_{total} increases, so K gets larger.

This is the quantitative reason behind the Le Chatelier rule you learned at AS: heating favours the endothermic direction.

Equilibria in industry: yield against rate

An industrial chemist wants a high yield *and* a fast rate at a sensible cost. Equilibrium and kinetics often pull in opposite directions, so the conditions chosen are a compromise.

	Haber process $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta H = -92$ kJ mol^{-1}	**Contact process** $2SO_2 + O_2 \rightleftharpoons 2SO_3$, $\Delta H = -197$ kJ mol^{-1}
Temperature	about 450 °C. Lower would give a bigger K and more ammonia, but the rate would be far too slow.	about 450 °C, for the same reason.
Pressure	about 200 atm. Higher pressure favours the side with fewer gas molecules (4 → 2) and speeds the reaction, but costs more to build and run safely.	1 to 2 atm. K is already so large that the yield is about 99% without high pressure, so the expense is not worth it.
Catalyst	iron: equilibrium reached faster, K unchanged.	vanadium(V) oxide, V_2O_5.
Other	ammonia is liquefied and removed, and unreacted gases are recycled.	SO_3 is removed by absorbing it, and unreacted gases pass over further catalyst beds.

EXAM TIP

When explaining a choice of conditions, give both sides: what the condition does to the **equilibrium yield** (through K or the position) and what it does to the **rate** and the **cost**. A catalyst helps the rate and lets a lower temperature be used, saving energy, but never improves the yield.

Quick checks

QUICK CHECK

What are the units of K_c for $2NO_2(g) \rightleftharpoons N_2O_4(g)$?

- mol dm^{-3}
- dm^3 mol^{-1}
- no units
- mol^2 dm^{-6}

$K_c = [N_2O_4]/[NO_2]^2$: one concentration on top and two underneath gives (mol dm^{-3})$^{-1}$ = dm^3 mol^{-1}. mol dm^{-3} is the unit for the reverse equation.

QUICK CHECK

Which is the correct K_p expression for $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$?

- $\dfrac{p_{CO}\, p_{H_2}}{p_C\, p_{H_2O}}$
- $\dfrac{p_{H_2O}}{p_{CO}\, p_{H_2}}$
- $\dfrac{p_{CO}\, p_{H_2}}{p_{H_2O}}$
- $p_{CO}\, p_{H_2}$

Carbon is a solid, so it does not appear; steam is a gas and does. Products go on top.

QUICK CHECK

A gas mixture contains 0.60 mol of A and 1.40 mol of B at a total pressure of 250 kPa. What is the partial pressure of A?

- 75 kPa
- 107 kPa
- 150 kPa
- 0.30 kPa

$x_A = 0.60 \div 2.00 = 0.30$, so $p_A = 0.30 \times 250 = 75$ kPa. 107 kPa comes from dividing by the moles of B instead of the total.

QUICK CHECK

For an exothermic equilibrium, what happens when the temperature is raised?

- K increases and the position moves right
- K is unchanged but the position moves left
- K decreases but the position does not change
- K decreases and the position moves left

Temperature is the only thing that changes K. For an exothermic reaction $\Delta S_{surroundings} = -\Delta H/T$ falls as T rises, so ΔS_{total} and K fall, and the mixture holds less product.

QUICK CHECK

The pressure on the equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is increased at constant temperature. What happens to K_p and to the amount of SO_3?

- K_p increases; more SO_3
- K_p unchanged; more SO_3
- K_p unchanged; less SO_3
- K_p decreases; more SO_3

K_p depends only on temperature. With 3 gas molecules on the left and 2 on the right, higher pressure shifts the position right, so the amount of SO_3 rises.

QUICK CHECK

A reaction has $K_c = 2 \times 10^{12}$ at 298 K. What can you say about it?

- It is fast
- Reactants predominate at equilibrium
- It goes essentially to completion, and ΔS_{total} is positive
- ΔS_{total} is negative

K above about 10^{10} means essentially complete reaction, and $\Delta S_{total} = R \ln K$ is positive. K says nothing about the rate: a reaction can have a huge K and still be very slow.

QUICK CHECK

0.40 mol of PCl_5 is heated in a 2.0 dm^3 flask: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. At equilibrium 0.10 mol of Cl_2 is present. What is K_c?

- 0.033 mol dm^{-3}
- 0.067 mol dm^{-3}
- 0.025 mol dm^{-3}
- 0.017 mol dm^{-3}

Equilibrium: PCl_5 0.30 mol, PCl_3 0.10 mol, Cl_2 0.10 mol. Concentrations 0.15, 0.050 and 0.050 mol dm^{-3}, so $K_c = (0.050 \times 0.050) \div 0.15 = 0.017$ mol dm^{-3}. Using moles without dividing by the volume gives 0.033.

QUICK CHECK

Why is the Haber process run at about 450 °C rather than at a lower temperature?

- K is larger at 450 °C
- A lower temperature would give a better yield but far too slow a rate
- The iron catalyst increases K at 450 °C
- Ammonia decomposes below 450 °C

The forward reaction is exothermic, so K is actually larger at lower temperatures. 450 °C is a compromise that gives an acceptable rate with a reasonable yield. A catalyst never changes K.